

BRUNEI INTL, Brunei Darussalam

WMO: 963150

Lat: 4.944N

Lon: 114.928E

Elev: 22

StdP: 101.06

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.9	23.2	20.7	15.4	30.1	21.2	15.9	29.3	8.3	28.5	7.4	28.3	2.1	190	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	7.3	33.8	25.8	33.1	26.0	32.8	26.0	27.7	31.1	27.4	30.8	27.1	30.6	3.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.9	22.6	29.4	26.2	21.7	28.9	26.2	21.6	28.8	88.3	31.4	87.0	31.3	85.8	30.9	32.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.8	6.0	5.3	DB	21.2	35.5	2.1	1.2	19.7	36.3	18.5	37.0	17.3	37.7	15.8	38.6	(4)
(5)			WB	20.7	28.9	1.9	1.0	19.3	29.6	18.2	30.2	17.1	30.7	15.7	31.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.0	27.4	27.5	28.1	28.5	28.7	28.4	28.2	28.3	28.1	27.8	27.8	27.6	(6)	
	DBStd	1.01	0.98	0.96	0.87	0.82	0.94	0.98	0.98	1.06	0.96	0.94	0.78	0.93	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6582	541	491	560	553	578	551	565	566	544	552	533	547	(10)	
	CDD18.3	3541	282	258	302	303	320	301	307	308	294	294	283	289	(11)	
	CDH23.3	37491	2797	2576	3209	3423	3646	3303	3288	3332	3144	2991	2867	2914	(12)	
	CDH26.7	13756	844	751	1117	1357	1495	1309	1294	1329	1206	1070	1003	980	(13)	
(14) Wind	WSAvg	2.4	2.6	2.8	2.5	2.3	2.2	2.2	2.2	2.4	2.3	2.3	2.3	2.4	(14)	
(15) Precipitation	PrecAvg	2922	270	148	129	185	233	206	222	232	262	318	339	363	(15)	
	PrecMax	3693	627	443	336	446	524	628	454	712	722	581	515	716	(16)	
	PrecMin	2249	2	8	0	51	68	68	47	18	64	29	146	94	(17)	
	PrecStd	398	175	123	85	98	115	110	105	151	139	111	101	156	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.2	32.6	33.3	33.9	34.1	34.0	34.1	34.2	33.8	33.2	33.0	32.9	(19)	
		MCWB	25.7	25.3	25.7	26.3	26.4	25.4	25.5	25.4	25.7	25.8	25.8	26.0	(20)	
	2%	DB	31.2	31.3	32.2	32.9	33.2	33.1	33.1	33.2	32.9	32.2	32.1	32.0	(21)	
		MCWB	25.8	25.8	26.0	26.5	26.5	25.8	25.8	25.7	25.8	26.0	26.2	26.0	(22)	
	5%	DB	30.9	30.8	31.5	32.1	32.8	32.4	32.3	32.6	32.1	31.9	31.4	31.2	(23)	
		MCWB	25.8	25.9	26.0	26.4	26.5	26.0	25.9	25.7	25.9	26.0	26.2	26.1	(24)	
	10%	DB	30.1	30.1	30.9	31.7	32.0	31.9	31.8	31.9	31.6	31.1	30.9	30.8	(25)	
		MCWB	25.7	25.7	26.0	26.4	26.4	26.0	25.9	25.8	26.0	26.0	26.1	26.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.1	27.1	27.4	27.7	28.1	27.7	27.7	27.7	27.7	27.4	27.4	27.3	(27)	
		MCDB	29.8	30.1	30.8	31.2	31.7	30.9	31.4	31.0	31.1	31.2	30.6	30.3	(28)	
	2%	WB	26.6	26.6	26.9	27.4	27.6	27.2	27.1	27.2	27.1	26.9	26.9	26.9	(29)	
		MCDB	29.4	29.5	30.1	30.9	31.2	30.7	30.8	30.6	30.7	30.4	30.3	29.9	(30)	
	5%	WB	26.3	26.2	26.6	27.1	27.2	26.8	26.7	26.7	26.7	26.5	26.6	26.6	(31)	
		MCDB	29.0	29.0	29.7	30.6	30.8	30.3	30.3	30.2	30.3	30.0	29.9	29.6	(32)	
	10%	WB	26.1	26.0	26.2	26.7	26.9	26.5	26.4	26.4	26.3	26.2	26.3	26.2	(33)	
		MCDB	28.8	28.7	29.2	30.1	30.5	30.0	29.9	29.9	29.9	29.6	29.5	29.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	5.8	6.5	7.1	7.3	7.3	7.5	7.5	7.4	7.0	6.9	6.5	(35)	
		MCDBR	6.7	6.4	7.2	7.5	7.9	8.0	8.2	8.3	8.0	7.8	7.4	7.2	(36)	
	5% WB	MCWBR	2.4	2.3	2.5	2.6	2.7	2.7	2.9	2.9	2.9	2.7	2.7	2.6	(37)	
		MCWBR	6.2	6.0	6.6	7.1	7.3	7.4	7.5	7.5	7.4	7.2	7.0	6.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.414	0.424	0.427	0.426	0.408	0.411	0.409	0.447	0.448	0.457	0.414	0.412	(40)	
		taud	2.502	2.449	2.441	2.442	2.506	2.505	2.505	2.374	2.369	2.352	2.493	2.511	(41)	
	Edn at Noon		907	904	898	883	879	865	871	853	866	863	900	903	(42)	
		Edh at Noon	110	118	120	117	107	105	106	124	127	129	110	108	(43)	
(44) All-Sky Solar Radiation	RadAvg		4.76	5.38	5.72	5.76	5.37	5.08	5.11	5.23	5.11	4.83	4.85	4.70	(44)	
	RadStd		0.44	0.40	0.32	0.23	0.23	0.23	0.24	0.20	0.30	0.32	0.29	0.34	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		+0.19	N/A	N/A	N/A	+0.17	+0.18	N/A	N/A	N/A	N/A	
(47)	Regional (3 neighbors)	+0.15	N/A	N/A	+0.29	+0.26	+0.23	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page